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On the construction and computation of Hochschild cohomology

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ABSTRACT

Hochschild cohomology is a powerful invariant of associative algebras, used in a variety of fields of mathematics and mathematical physics. It is, however, often difficult to compute Hochschild cohomology explicitly. In this thesis we present its construction along with some computations. Hochschild cohomology is introduced through the classical bar complex. We show that, when working over a field, the bar complex is a projective resolution. This leads to the Hochschild cohomology being equivalent to the Ext-groups over the enveloping algebra. Thus, any projective resolution may be used in the bar complex's place. One such resolution, the Bardzell resolution, is presented in the quadratic monomial case. It is proven to be a projective resolution, and examples are computed to illustrate its usefulness. This project provides the foundation for a master's thesis aiming to investigate the Hochschild cohomology of almost gentle algebras, a generalization of gentle algebras for which the Hochschild cohomology was recently described explicitly. This will be done using the Bardzell resolution as constructed in this thesis.

SAMMENDRAG

Hochschild-kohomologi er en kraftig invariant av assosiative algebraer, brukt i en rekke felt innen matematikk og matematisk fysikk. Det er imidlertid ofte vanskelig å beregne Hochschild-kohomologi eksplisitt. I denne oppgaven presenterer vi dens konstruksjon sammen med noen beregninger. Hochschild-kohomologi introduseres gjennom det klassiske barkomplekset. Vi viser at når man arbeider over et felt, er barkomplekset en projektiv oppløsning. Dette fører til at Hochschild-kohomologien er ekvivalent med Ext-gruppene over innhylningsalgebraen. Dermed kan enhver projektiv oppløsning brukes i stedet for barkomplekset. En slik oppløsning, Bardzell-oppløsningen, presenteres i det kvadratiske monomielle tilfellet. Vi beviser at det er en projektiv oppløsning, og beregner eksempler for å illustrere dens nytteverdi. Dette prosjektet danner grunnlaget for en masteroppgave som tar sikte på å undersøke Hochschild-kohomologien til nesten milde algebraer, en generalisering av milde algebraer der Hochschild-kohomologien nylig ble beskrevet eksplisitt. Dette vil bli gjort ved hjelp av Bardzell-oppløsningen slik den er konstruert i denne oppgaven.

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CONVENTION AND NOTATION

We will use the following notation and conventions throughout this thesis:

- We let $\Bbbk$ denote a field.
- We let A denote a $\Bbbk$-algebra.
- For a ring R, we let ${}_R\mathrm{Mod}$ and Mod_R denote the categories of left and right R-modules respectively. Given a second ring S, we let ${}_R\mathrm{Mod}_S$ denote the category of R-S-bimodules.
- When we refer to a module, without specifying if it is a left module, right module or bimodule, we always consider it a left module.
- We denote a quiver $Q = (Q_0, Q_1, s, t)$, where Q_0 is the set of vertices, Q_1 is the set of arrows and s and t the maps sending an arrow to its source and target respectively.
- Given the quiver $1 \xrightarrow{\alpha} 2 \xrightarrow{\beta} 3$ we write the composition of the two arrows $\alpha\beta$.
- Given two composable morphisms f and g, we denote $f \circ g$ by fg.

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1 Introduction

Hochschild cohomology was first introduced by its namesake Gerhard Hochschild in 1945 [5]. It is a powerful invariant of algebras. It is invariant under Morita equivalence, but more strongly it is invariant under derived equivalence, as shown by Rickard in 1991 [9]. Along with Hochschild homology and the later introduced Gerstenhaber bracket, Connes differential, and the cup and cap product, it forms what is called the Tamarkin-Tsygan calculus [3]. This carries significant amounts of information. We will focus solely on the Hochschild cohomology in this thesis.

Applications of Hochschild cohomology span multiple fields of mathematics, including topology, algebraic geometry and representation theory, see e.g. [7]. Moreover, it has uses within mathematical physics, including string theory, where the fact that the second Hochschild cohomology group classifies the infinitesimal deformations of an algebra [12] has been utilized. This allowed an analysis of the relation between open and closed strings [6]. It follows that the ability to compute these cohomology groups explicitly is of importance. However, we will see that this is not as straightforward as one might wish.

Hochschild originally defined Hochschild cohomology through the bar complex, and we follow this approach in this thesis. While good for theoretical considerations, the bar complex quickly grows large and is not well-suited for computation. It is therefore common to use alternative complexes whenever possible. The Hochschild cohomology groups and related computations are in fact so intricate that, to our knowledge, there is only one complete class of algebras which has had its Tamarkin-Tsygan calculus fully described. This is the class of gentle algebras and was done by Chaparro et al. in 2024 [3]. Gentle algebras are interesting as they are simple enough to make explicit computations manageable, but complex enough to have rich theory. For more on gentle algebras see [4].

Understanding these results for gentle algebras is particularly interesting in light of the notion of almost gentle algebras introduced by Green and Schroll in 2017 [4], which is a vast generalization of gentle algebras. It is natural and interesting to ask whether the description of the Hochschild cohomology for gentle algebras extends to this setup. While being a much more general class of algebras, certain qualities of gentle algebras remain. Importantly, both gentle and almost gentle algebras are quadratic monomial [4]. A key difference, however, is that while gentle algebras are all of tame representation type, almost gentle algebras can be wild. Extending a method applied for tame algebras to a class with wild algebras would be a very interesting result. Investigating the Hochschild cohomology of the gentle and almost gentle algebras will be the subject of the master's thesis building on this project. It is therefore worthwhile to become familiar with the subject of Hochschild cohomology.

The purpose of this thesis is exactly this. We aim to present the construction and computation of Hochschild cohomology in a concrete and approachable way. We conclude that any projective resolution may be used when we are working over a field, and introduce an alternative resolution to the bar complex. This alternative resolution, the Bardzell resolution, has some very desirable properties [2]. Importantly, it stops where the bar complex does not. While it is originally defined for

all monomial algebras, we only consider the case for quadratic monomial algebras, which gives a particularly simple resolution.

This thesis assumes familiarity with the basic theory of rings and modules. We will mostly discuss subjects within the field of homological algebra. Therefore, some key concepts are recalled, namely tensor products, complexes, projectivity and cohomology. To fully follow the computations and results in this thesis, some familiarity with category theory and representation theory is recommended, although an introduction to quivers and path algebras will also be presented. We start the thesis with Section 2, briefly covering the aforementioned subjects, before we move on to the classical construction of Hochschild cohomology via the bar complex in Section 3. Finally, we define and perform some computations with the Bardzell resolution in Section 4. This is the resolution which will be applied further in the master's thesis.

2 Background

To be able to obtain a good understanding of Hochschild cohomology, there are some background topics that should be properly understood. As stated in the introduction, we will therefore recall some concepts in particular. We consider the tensor product, complexes, projectivity and cohomology. This is because these notions play a key role in the construction of Hochschild cohomology. We will also briefly cover quivers and path algebras as Section 4 relies on these notions.

2.1 Tensor products

Tensor products allow us, in a sense, to combine two modules into a new one, along with some useful properties. It is central to the historical definition of Hochschild cohomology. This subsection is based on [8] and [10]. Let R be a ring, M a right R-module, and N a left R-module. We aim to produce $M \otimes_R N$, the tensor product of M and N over R. First, we define a property we wish to have for the tensor product.

Definition 1. Let H be an abelian group. A map $f : M \times N \longrightarrow H$ is called *R-biadditive*, if for any $m, \tilde{m} \in M$, $n, \tilde{n} \in N$ and $r \in R$, we have

$$\begin{aligned} f(m + \tilde{m}, n) &= f(m, n) + f(\tilde{m}, n), \\ f(m, n + \tilde{n}) &= f(m, n) + f(m, \tilde{n}), \\ f(mr, n) &= f(m, rn). \end{aligned}$$

Note that an R-biadditive map is also called R-balanced. We are now ready to define the tensor product.

Definition 2. The *tensor product* of M and N over R is an abelian group $M \otimes_R N$, along with an R-biadditive map $f : M \times N \longrightarrow M \otimes_R N$. The tensor product must satisfy the following universal property: Given any abelian group H and any

R-biadditive map $h : M \times N \longrightarrow H$, there must exist a unique morphism of abelian groups $\tilde{h} : M \otimes_R N \longrightarrow H$ such that the diagram

$$\begin{tikzcd} M \times N \arrow[rr, "f"] \arrow[dr, "h"'] & & M \otimes_R N \arrow[dl, dashed, "\tilde{h}"] \\ & H & \end{tikzcd}$$

commutes.

The universal property ensures that any other candidate for the tensor product must factor uniquely through it. It remains to show that this construction actually exists.

Theorem 3. *The tensor product of M and N over R exists and is unique up to unique isomorphism.*

Proof. To show that the tensor product exists, we will construct a candidate and show that it exhibits the universal property. Let G be the free abelian group generated by the cartesian product $M \times N$. Let U be the subgroup of G generated by all elements of the form

$$\begin{gathered} (m + \tilde{m}, n) - (m, n) - (\tilde{m}, n), \\ (m, n + \tilde{n}) - (m, n) - (m, \tilde{n}), \\ (mr, n) - (m, rn), \end{gathered}$$

for $m, \tilde{m} \in M$, $n, \tilde{n} \in N$ and $r \in R$. We will now consider the abelian group G/U along with the map

$$f : M \times N \longrightarrow G/U, \ f(m, n) = (m, n) + U$$

as a candidate for the tensor product. We need to show that f is in fact R-biadditive and that the universal property holds. Observe that f respects the first requirement for being R-biadditive as for $m, \tilde{m} \in M$ and $n \in N$, we have

$$\begin{aligned} f(m + \tilde{m}, n) &= (m + \tilde{m}, n) + U \\ &= (m, n) + (\tilde{m}, n) + (m + \tilde{m}, n) - (m, n) - (\tilde{m}, n) + U \\ &= (m, n) + (\tilde{m}, n) + U = f(m, n) + f(\tilde{m}, n). \end{aligned}$$

Here we use that $(m + \tilde{m}, n) - (m, n) - (\tilde{m}, n)$ is an element of U by definition. By analogous arguments, the remaining two requirements follow.

Now to show the universal property, we introduce an abelian group H and an R-biadditive map $h : M \times N \longrightarrow H$. Consider the solid part of the diagram

$$\begin{tikzcd} M \times N \arrow[rr, "f"] \arrow[dr, "i"] \arrow[ddr, "h"'] & & G/U \\ & U \arrow[d, "j"] & \\ & G \arrow[uur, "\pi"] \arrow[d, dashed, "\varphi"'] & \\ & H \arrow[uuur, dashed, "\tilde{h}"] & \end{tikzcd}$$

where i and j are inclusions, and π is the natural projection. Clearly, we have $\pi i = f$, so the upper triangle commutes. Because G is free, and h is defined on the basis, we can extend h to a morphism from G. We call this morphism φ, as represented by the vertical dashed arrow in the above diagram. We now have $\varphi i = h$. Since h is R-biadditive, so is φ.

Notice that G/U is the cokernel of the inclusion $j : U \longrightarrow G$, meaning that any morphism which postcomposes with j and returns 0, factors uniquely through π. Since φ is R-biadditive, $U \subseteq \ker(\varphi)$. Hence, we have a unique $\tilde{h} : G/U \longrightarrow H$ such that $\varphi = \tilde{h}\pi$ and we can conclude

$$h = \varphi i = \tilde{h}\pi i = \tilde{h} f.$$

As stated $\tilde{h}$ is unique as it arises from the universal property of the cokernel, and therefore our candidate fulfills the universal property of the tensor product. Uniqueness of the tensor product follows naturally from the universal property. $\square$

Using the above notation we let $M \otimes_R N = G/U$. For $m \in M$ and $n \in N$, $m \otimes n = f(m, n)$, and is called an elementary tensor. Note that because of G's construction, all elements in $M \otimes_R N$ are finite sums of elementary tensors. There are some computational properties worth noting that follow from the definition of the tensor product. For $m, \tilde{m} \in M$, $n, \tilde{n} \in N$ and $r \in R$ the following holds:

- $mr \otimes n = m \otimes rn$,
- $(m + \tilde{m}) \otimes n = m \otimes n + \tilde{m} \otimes n$,
- $m \otimes (n + \tilde{n}) = m \otimes n + m \otimes \tilde{n}$

We now compute two tensor products for intuition.

Example 4.

1. First we study the tensor product of two $\mathbb{Z}$-modules, namely $\mathbb{Z}/k\mathbb{Z}$ and $\mathbb{Z}/l\mathbb{Z}$, for two integers k, l. The aim is to compute $\mathbb{Z}/k\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/l\mathbb{Z}$. We claim that $\mathbb{Z}/k\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/l\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z}$, where d is the greatest common divisor of l and k. To see this consider the diagram

$$\begin{tikzcd}
\mathbb{Z}/k\mathbb{Z} \times \mathbb{Z}/l\mathbb{Z} \arrow[rr, "f"] \arrow[dr, "h"'] & & \mathbb{Z}/k\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/l\mathbb{Z}. \arrow[dl, dashed, "\tilde{h}"] \\
& \mathbb{Z}/d\mathbb{Z} &
\end{tikzcd}$$

where $h : \mathbb{Z}/k\mathbb{Z} \times \mathbb{Z}/l\mathbb{Z} \longrightarrow \mathbb{Z}/d\mathbb{Z}$ is defined by $h(z + k\mathbb{Z}, z' + l\mathbb{Z}) = zz' + d\mathbb{Z}$, for any $(z + k\mathbb{Z}, z' + l\mathbb{Z}) \in \mathbb{Z}/k\mathbb{Z} \times \mathbb{Z}/l\mathbb{Z}$. The map h is $\mathbb{Z}$-biadditive and well-defined. Thus, by the universal property of the tensor product, $\tilde{h}$ must exist, making the diagram commute. Since the diagram commutes, it is clear that

$$\tilde{h}(z + k\mathbb{Z} \otimes z' + l\mathbb{Z}) = zz' + d\mathbb{Z}.$$

This is an isomorphism, as seen by constructing the inverse

$$\tilde{h}^{-1} : \mathbb{Z}/d\mathbb{Z} \longrightarrow \mathbb{Z}/k\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/l\mathbb{Z},$$

by $\tilde{h}^{-1}(z + d\mathbb{Z}) = z \otimes 1$. Clearly, $\tilde{h}^{-1}$ is well-defined, and both precompose and postcompose to the identity with $\tilde{h}$. This can be seen using the above mentioned computational properties of the tensor product. Thus, the tensor product $\mathbb{Z}/k\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/l\mathbb{Z}$ is in fact isomorphic to $\mathbb{Z}/d\mathbb{Z}$.

2. Now consider $R \otimes_R N$ for some left R-module N. Notice that the tensor product is over the ring R and we consider R as a right module over itself. Note that for an arbitrary elementary tensor $r \otimes n \in R \otimes_R N$, we have

$$r \otimes n = 1 \cdot r \otimes n = 1 \otimes rn.$$

Because the tensor product is over R, the tensor factor from R can always be multiplied into the tensor factor from N using the module structure. With the map $r \otimes n \mapsto rn$ and its inverse $n \mapsto 1 \otimes n$, it follows immediately that the tensor product is isomorphic to N along with the R-biadditive map $(r, n) \mapsto rn$.

2.2 Complexes

The goal of this subsection is to define complexes, which is a necessary step in the definition of homology, and we base this subsection on [10]. Complexes are one of the fundamental building blocks of homological algebra, as it gives rise to the notions of exactness, (co)homology and other central topics within the field.

Definition 5. A (cochain) *complex* in an abelian category $\mathcal{A}$ is a sequence

$$(X^\bullet, d^\bullet) = \big((X^n)^{n\in\mathbb{Z}}, (d^n : X^n \longrightarrow X^{n+1})_{n\in\mathbb{Z}}\big),$$

where all $X^n \in \mathcal{A}$ and $d^n \in \mathrm{Hom}_{\mathcal{A}}(X^n, X^{n+1})$, with the property that $d^{n+1}d^n = 0$ for any $n \in \mathbb{Z}$.

We say X^n is in degree n of the complex, and call d^n the differentials of the complex. Using the notation of the above definition an arbitrary complex can be visualized as

$$\cdots \xrightarrow{d^{n-2}} X^{n-1} \xrightarrow{d^{n-1}} X^n \xrightarrow{d^n} X^{n+1} \xrightarrow{d^{n+1}} \cdots$$

Remark 6. In Definition 5, the definition of a cochain complex is given. One can also consider a chain complex, which we denote by

$$(X_\bullet, d_\bullet) = ((X_n)_{n\in\mathbb{Z}}, (d_n : X_n \longrightarrow X_{n-1})_{n\in\mathbb{Z}}.$$

The only difference being the use of lower indices which decrease instead of increase as is the case for cochain complexes. The complex-requirement therefore becomes $d_{n-1}d_n = 0$. Both cochain and chain complexes will be used in this thesis.

We will now present two simple examples of complexes.

Example 7.

1. Let H be a subgroup of an abelian group G. We can then construct

$$\cdots \xrightarrow{0} 0 \xrightarrow{0} H \xrightarrow{i} G \xrightarrow{\pi} G/H \xrightarrow{0} 0 \xrightarrow{0} \cdots$$

   Here, $i : H \longrightarrow G$ is the inclusion and $\pi : G \longrightarrow G/H$ is the natural projection. To see that the adjacent arrows compose to zero, it suffices to check that $\pi i = 0$. Observe that H along with the inclusion i is *exactly* the kernel of π, so they must compose to zero.

2. We construct a second sequence of abelian groups. Consider

$$\cdots \xrightarrow{0} \mathbb{Z} \xrightarrow{-\cdot d} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{-\cdot d} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{-\cdot d} \cdots$$

   where $- \cdot d$ is simply multiplication by some d in $\mathbb{Z}$. Notice that composing adjacent arrows clearly results in zero.

We briefly restrict our attention to module categories, as that is all that is needed in this thesis. The complex property of a sequence $(X^\bullet, d^\bullet)$ in a module category, implies that $\mathrm{im}(d^{n-1}) \subseteq \ker(d^n)$ for all $n \in \mathbb{Z}$. If the opposite inclusion also holds, we say the complex is *exact*.

Definition 8. Let R be a ring. A sequence $(X^\bullet, d^\bullet)$ in ${}_R\mathrm{Mod}$ is *exact* in degree n if $\ker(d^n) = \mathrm{im}(d^{n-1})$. We say the sequence is *exact* if it is exact in every degree.

Remark 9. Considering the sequences in Example 7 as sequences of $\mathbb{Z}$-modules, we can see that the first complex is exact, while the second is not.

2.3 Projective resolutions

The projective resolution of an object is a complex which carries some information about the original object. While projective resolutions are not unique, certain constructions do not depend on the choice of projective resolution. In this subsection we aim to define and build intuition around the notion of a projective object and a projective resolution. This subsection is based on [8] and [10].

Before we can consider projective resolutions, we must define what a projective object is.

Definition 10. Let $\mathcal{A}$ be an abelian category. An object $P \in \mathcal{A}$ is *projective* if for any objects $C, D \in \mathcal{A}$ and for any morphism $p : P \longrightarrow D$, and epimorphism $f : C \longrightarrow D$, there exists a morphism $h : P \longrightarrow C$ such that $fh = p$. This can be visualized as

$$\begin{tikzcd} & P \arrow[dl, dashed, "h"'] \arrow[d, "p"] \\ C \arrow[r, two heads, "f"'] & D \end{tikzcd}$$

where the solid part of the diagram must imply the existence of the dashed morphism making the diagram commute.

Remark 11. The given definition of projective objects is equivalent to stating that $P \in \mathcal{A}$ is projective if and only if $\mathrm{Hom}_{\mathcal{A}}(P, -) : \mathcal{A} \longrightarrow \mathbf{Ab}$ is exact as a functor.

Projective elements are not always the easiest to identify directly from the definition. We will now present a simple way to identify projective modules.

Example 12. Let R be a ring. In this example we will show that all free R-modules are projective objects in the category ${}_R\mathrm{Mod}$. Note that the result holds in Mod_R as well for analogous reasons.

Let P be such a free R-module with basis X. We then consider two R-modules $C, D \in {}_R\mathrm{Mod}$ along with an epimorphism $f \in \mathrm{Hom}_{{}_R\mathrm{Mod}}(C, D)$, and a morphism $p \in \mathrm{Hom}_{{}_R\mathrm{Mod}}(P, D)$. The resulting diagram looks exactly as the solid part of the one in Definition 10. We must now show the existence of the dashed morphism. Since f is an epimorphism in ${}_R\mathrm{Mod}$, it is surjective. Therefore, for every basis element $x \in X$, $p(x) = f(c_x)$ for some $c_x \in C$. By the Axiom of Choice, we can now define a morphism $h' : X \longrightarrow C$ on the basis X by

$$h'(x) = c_x, \text{ for all } x \in X,$$

where c_x is one such element in C such that $p(x) = f(c_x)$. One can extend h' onto all of P, using that any $y \in P$ can be written as a finite sum $\sum_{x \in X} r_x x$, where $r_x \in R$. This results in a morphism h acting on y by

$$h(y) = h(\sum_{x \in X} r_x x) = \sum_{x \in X} r_x h'(x).$$

Postcomposing with f yields

$$fh(y) = \sum_{x \in X} r_x f(c_x) = \sum_{x \in X} r_x p(x) = p\big(\sum_{x \in X} r_x x\big) = p(y).$$

Hence, P is projective by Definition 10.

The result that free modules are projective is very useful, as it gives us an alternative way to identify projectives. It so happens, that not only are all free modules projective, all projective modules are direct summands of free modules, for a proof of this see, e.g. Theorem 3.5 in [10]. Given the notion of a projective object we can now define a projective resolution.

Definition 13. Let $\mathcal{A}$ be an abelian category. A *projective resolution* of some $X \in \mathcal{A}$ is a complex

$$\cdots \xrightarrow{p_3} P_2 \xrightarrow{p_2} P_1 \xrightarrow{p_1} P_0 \longrightarrow 0,$$

in $\mathcal{A}$, where all P_i are projective and the sequence is exact everywhere, except in degree 0, in which the cokernel of p_1 is X.

A natural question to ask when introducing a new definition is the question of existence. In some cases this is trivial, but in this case it is not. To guarantee a projective resolution of any object, we need to be operating in a category with ample projective objects to take from. Such a category is said to have *enough projectives*.

Definition 14. An abelian category $\mathcal{A}$ has *enough projectives* if for all objects in $\mathcal{A}$ there exists an epimorphism to said object from a projective object in $\mathcal{A}$.

The notion of having enough projectives allows us to iteratively construct a projective resolution of any object in the category. Note that projective resolutions can exist in categories which does not have enough projectives, but they are not guaranteed to exist for every object.

2.4 (Co)homology

We are now ready to define one of the central notions of homological algebra, namely homology. We only consider homology of complexes in module categories. This subsection follows [10] and [8]. We work with cochain complexes in this subsection, and therefore only discuss cohomology.

Given a ring R, let $X^\bullet$ be a complex in ${}_R\mathrm{Mod}$, as follows

$$\cdots \xrightarrow{d^{n-2}} X^{n-1} \xrightarrow{d^{n-1}} X^n \xrightarrow{d^n} X^{n+1} \xrightarrow{d^{n+1}} \cdots, \tag{2.4.1}$$

with X^n being in degree n. The cohomology of this sequence can be said to measure *how far away* this sequence is from being exact.

Definition 15. The *n-th cohomology* of the sequence (2.4.1) is defined as

$$H^n(X^\bullet) = \ker(d^n)/\mathrm{im}(d^{n-1}).$$

Notice that both $\ker(d^n)$ and $\mathrm{im}(d^{n-1})$ are submodules of X^n, particularly $\mathrm{im}(d^{n-1})$ must be a submodule of $\ker(d^n)$ as a consequence of the complex property. If $X^\bullet$ were exact in degree n, then the cohomology group would be 0 in degree n, as the kernel of d^n would be isomorphic to the image of d^{n-1}. Therefore, the more non-zero cohomology groups there exist of a sequence, the further it can be said to be from being exact. For intuition, we compute the following simple example.

Example 16. Consider the following sequence in ${}_\mathbb{Z}\mathrm{Mod}$,

$$\cdots \xrightarrow{0} 0 \xrightarrow{0} \mathbb{Z}/k\mathbb{Z} \xrightarrow{-\cdot l} \mathbb{Z}/k\mathbb{Z} \xrightarrow{0} 0 \xrightarrow{0} \cdots \tag{2.4.2}$$

with $- \cdot l$ being multiplication by l, for some $k, l \in \mathbb{Z}$. We define the first non-zero entry to be in degree 1. The cohomology groups are

$$H^0 = \ker(0)/\mathrm{im}(0) = 0/0 = 0$$

$$H^1 = \ker(- \cdot l)/\mathrm{im}(0) = \ker(- \cdot l) \cong \mathbb{Z}/d\mathbb{Z}$$

$$H^2 = \ker(0)/\mathrm{im}(- \cdot l) \cong \mathbb{Z}/d\mathbb{Z}$$
$$H^3 = 0/0 = 0,$$

where d is the greatest common divisor of k and l. Notice that all other cohomology groups are necessarily zero.

Remark 17. Given some chain complex $(X_\bullet, d_\bullet)$, the definition of the n-th homology, $H_n(X_\bullet)$, is completely analogous, simply with the different indexing of a chain complex instead of a cochain complex.

2.5 Quivers and path algebras

In Section 4 we need familiarity with path algebras and quivers to construct the so-called Bardzell resolution. We therefore introduce some notation and definition. This subsection is based on [1], [3] and [12]. We define a quiver Q as a quadruple $Q = (Q_0, Q_1, s, t)$, where

- Q_0 is the set of *vertices*,
- Q_1 is the set of *arrows*,
- $s : Q_1 \longrightarrow Q_0$ is the map taking an arrow to its *source*,
- $t : Q_1 \longrightarrow Q_0$ is the map taking an arrow to its *target*.

A path p in Q is denoted by $p = p_1 p_2 \cdots p_m$, where $p_i \in Q_1$ are arrows for all i. We say p has length m, i.e. the number of arrows it is composed of. Every vertex in Q_0 represents a path of length 0, by standing still at this vertex. A path q is called a subpath of p if $p = xqz$ for some paths x, z in Q. Note that x and z can be paths of length 0.

Let $p = p_1 p_2 \cdots p_m$ and $q = q_1 q_2 \cdots q_k$ be two paths in Q of length m and k, respectively. We let

$$pq = \begin{cases} p_1 p_2 \cdots p_m q_1 q_2 \cdots q_k, & t(p_m) = s(q_1) \\ 0, & t(p_m) \neq s(q_1) \end{cases}$$

Below we present two simple examples of quivers. Notice that in the left case, the arrow, x, can be repeated any number of times. However, this is clearly not the case for any arrow in the right diagram.

$$\begin{tikzcd}
\bullet \arrow[loop right, "x"] & & & & & & \\
& 1 \arrow[r, "a"] & 2 \arrow[r, "b"] \arrow[dr, "d"'] & 3 \arrow[dr, "c"] & & & \\
& & & & 5 \arrow[r, "f"] & 6 \\
& & & 4 \arrow[ur, "e"'] & & &
\end{tikzcd}$$

Given a field $\Bbbk$, we call the $\Bbbk$-algebra, whose underlying $\Bbbk$-vector space has basis consisting of all paths in Q, the *path algebra* of Q denoted by $\Bbbk Q$. Multiplication in the algebra is defined by the above defined concatenation of paths in Q. Notice that we can take quotients of path algebras. For example consider a set of paths P, we can consider the ideal $I = \Bbbk P$, the $\Bbbk$-vector space spanned by P. Then $\Bbbk Q/I$ is a possible algebra to consider.

Example 18. Denote the left case of the two above quivers by Q. The path algebra $\Bbbk Q$ has basis $\{e, x, x^2, x^3 \dots\}$, where e represents the arrow of length zero at the single vertex. This path algebra is in fact isomorphic to $\Bbbk[x]$, simply by sending e to 1, x to x, x^2 to x^2, and so on. If we introduce an ideal $I = (x^n)$, we can consider $\Bbbk Q/I$, which is then isomorphic to $\Bbbk[x]/(x^n)$ by the same argument.

3 Hochschild cohomology

We move on to introduce Hochschild cohomology. This will be done in detail, before we present some examples of the computation. This section is based on the textbook Hochschild Cohomology for Algebras by Sarah J. Witherspoon [12]. Throughout this section let $\Bbbk$ be a field and A be a $\Bbbk$-algebra. All tensor products in this section are taken over $\Bbbk$. We will define Hochschild cohomology through the bar complex similarly to how Hochschild defined it in 1945 [5]. To do this, we must first introduce the so-called enveloping algebra.

3.1 The enveloping algebra

The enveloping algebra is a construction that is convenient to use for the bar complex and Hochschild cohomology. Given our $\Bbbk$-algebra A, we define the *opposite algebra* A^{op} by having the same elements as A, but with multiplication defined by

$$a_1 \cdot_{\mathrm{op}} a_2 = a_2 \cdot a_1,$$

for all $a_1, a_2 \in A^{\mathrm{op}}$. Here, the multiplication on the right-hand side is done in A.

The opposite algebra allows us to define the *enveloping algebra*,

$$A^{\mathrm{e}} = A \otimes_{\Bbbk} A^{\mathrm{op}}$$

with multiplication

$$(a_1 \otimes a_2) \cdot (\tilde{a}_1 \otimes \tilde{a}_2) = a_1\tilde{a}_1 \otimes \tilde{a}_2 a_2,$$

for $a_1, a_2, \tilde{a}_1, \tilde{a}_2 \in A$. Notice that, technically, a_2 and $\tilde{a_2}$ are in A^{op}, resulting in the multiplication being reversed in the second tensor factor. The enveloping algebra is, as the name suggests, an algebra and we can therefore consider its modules. We define a left A^{e}-module M, through the module action

$$(a_1 \otimes a_2)m = a_1 m a_2,$$

for $a_1, a_2 \in A$ and $m \in M$. Notice that since the tensor product is over $\Bbbk$, it follows that

$$km = (k \otimes 1)m = (1 \otimes k)m = mk,$$

for $k \in \Bbbk$ and $m \in M$. When we refer to an A^{e}-module in this thesis, we always mean a left A^{e}-module.

We consider an A-bimodule, N, as a $\Bbbk$-module which is also both a left and right A-module, such that

$$(a_1 n)a_2 = a_1(na_2),$$

for $a_1, a_2 \in A, n \in N$. We also require that the left and right $\Bbbk$-action agrees, i.e. $kn = nk$ for all $k \in \Bbbk$ and $n \in N$. Observe that N being an A-bimodule is equivalent to N being a left A^{e}-module, as the module actions coincide. For this reason, the two terms will be used interchangeably throughout the remaining thesis.

Importantly, the tensor product $A^{\otimes n+2} = \overbrace{A \otimes_{\Bbbk} \cdots \otimes_{\Bbbk} A}^{n+2 \text{ times}}$, for $n \geq 0$ is an A^e-module (or A-bimodule) by

$$(a_0 \otimes a_1) \cdot (\tilde{a}_0 \otimes \cdots \otimes \tilde{a}_{n+1}) = a_0\tilde{a}_0 \otimes \tilde{a}_2 \otimes \cdots \otimes \tilde{a}_n \otimes \tilde{a}_{n+1}a_1.$$

Here, $a_0 \otimes a_1 \in A^{\mathrm{e}}$ and $\tilde{a}_0 \otimes \cdots \otimes \tilde{a}_{n+1} \in A^{\otimes n+2}$. Notice that only the first and last tensor factor is affected by the module action. As a consequence of this, observe that $f \in \mathrm{Hom}_{A^{\mathrm{e}}}(A^{\otimes n+2}, M)$ is A^{e}-linear if

$$f(a_0 \otimes a_1 \otimes \cdots \otimes a_n \otimes a_{n+1}) = (a_0 \otimes a_{n+1})f(1 \otimes a_1 \otimes \cdots \otimes a_n \otimes 1),$$

for $a_0 \otimes \cdots \otimes a_{n+1} \in A^{n+2}$. Thus, in a sense, an A^{e}-linear morphism from $A^{\otimes n+2}$ is decided by its action on the middle n tensor factors. This is simply the definition of linearity with respect to the module action, but it is worth emphasizing as it will be important to keep in mind in the rest of the thesis.

3.2 The bar complex

We will now construct the bar complex and consider it as a complex of modules over the enveloping algebra or equivalently, as a complex of A-bimodules. Later the Hochschild cohomology will arise from this complex. Notice that while the definition is not too cumbersome, the resolution quickly grows big and ends up being poorly suited for computation in higher degrees. It is, however, good for intuition and theory. After defining the bar complex, the aim is to prove Theorem 25. This will be done by breaking up the result into multiple lemmas.

Definition 19. The *bar complex* of A is

$$B(A) : \cdots \xrightarrow{d_3} A^{\otimes 4} \xrightarrow{d_2} A^{\otimes 3} \xrightarrow{d_1} A^{\otimes 2} \xrightarrow{0} 0$$

We think of $B(A)$ as a chain complex with $B(A)_n = A^{\otimes n+2}$. Let

$$\delta_n^i : A^{\otimes n+2} \longrightarrow A^{\otimes n+1}$$

be the map multiplying the i-th tensor factor with the $(i+1)$-th, i.e.

$$\delta_n^i(a_0 \otimes \cdots \otimes a_{n+1}) = a_0 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_{n+1}.$$

We then define the differential d_n of $B(A)$ by

$$d_n = \sum_{i=0}^{n} (-1)^i \delta_n^i.$$

As seen in the last section, all objects in the bar complex are A^e-modules, or equivalently A-bimodules. It is also clear that the differentials are A^e-linear, so $B(A)$ can be considered as a sequence of A^e-modules. To build some intuition concerning the differentials, we compute d_1 and d_2.

Example 20. Let $a_i \in A$ for integer indexes i. Applying d_1 to an arbitrary elementary tensor $a_0 \otimes a_1 \otimes a_2$ yields

$$d_1(a_0 \otimes a_1 \otimes a_2) = \delta_1^0(a_0 \otimes a_1 \otimes a_2) - \delta_1^1(a_0 \otimes a_1 \otimes a_2) = (a_0 a_1 \otimes a_2) - (a_0 \otimes a_1 a_2).$$

Similarly, computing d_2 results in

$$\begin{aligned} d_2(a_0 \otimes a_1 \otimes a_2 \otimes a_3) &= \delta_2^0(a_0 \otimes a_1 \otimes a_2 \otimes a_3) - \delta_2^1(a_0 \otimes a_1 \otimes a_2 \otimes a_3) \\ &\quad + \delta_2^2(a_0 \otimes a_1 \otimes a_2 \otimes a_3) \\ &= (a_0 a_1 \otimes a_2 \otimes a_3) - (a_0 \otimes a_1 a_2 \otimes a_3) + (a_0 \otimes a_1 \otimes a_2 a_3). \end{aligned}$$

We now compute the composition, showing that $d_1 d_2 = 0$, which is a special case of Lemma 21. We get

$$\begin{aligned} d_1 d_2(a_0 \otimes a_1 \otimes a_2 \otimes a_3) &= d_1((a_0 a_1 \otimes a_2 \otimes a_3) - (a_0 \otimes a_1 a_2 \otimes a_3) + (a_0 \otimes a_1 \otimes a_2 a_3)) \\ &= a_0 a_1 a_2 \otimes a_3 - a_0 a_1 \otimes a_2 a_3 - (a_0 a_1 a_2 \otimes a_3 - a_0 \otimes a_1 a_2 a_3) \\ &\quad + (a_0 a_1 \otimes a_2 a_3 - a_0 \otimes a_1 a_2 a_3) \\ &= 0. \end{aligned}$$

As we can see, the composition is zero. Note from the computations above that this holds because $\delta_1^0 \delta_2^0 = \delta_1^0 \delta_2^1$, $\delta_1^1 \delta_2^0 = \delta_1^0 \delta_2^2$ and $\delta_1^1 \delta_2^1 = \delta_1^1 \delta_2^2$.

We now extend the observation above and prove that the bar complex is in fact a complex.

Lemma 21. *The bar complex $B(A)$ is a complex of A^e-modules.*

Proof. We want to show that, for any integer $n \geq 2$, we have $\mathrm{im}(d_n) \subseteq \ker(d_{n-1})$, that is, $d_{n-1} d_n = 0$. As seen in Example 20, this holds for $n = 2$. For any $n \geq 2$, consider

$$d_{n-1} d_n = d_{n-1}(\sum_{i=0}^{n} (-1)^i \delta_n^i) = \sum_{j=0}^{n-1} (-1)^j \delta_{n-1}^j \sum_{i=0}^{n} (-1)^i \delta_n^i = \sum_{j=0}^{n-1} \sum_{i=0}^{n} (-1)^{i+j} \delta_{n-1}^j \delta_n^i.$$

Observe next that $\delta_{n-1}^j \delta_n^i = \delta_{n-1}^i \delta_n^{j+1}$ for $i \leq j$. We divide the sum above into two cases, $i \leq j$ and $j < i$, and apply this observation. This yields

$$\begin{aligned}
d_{n-1}d_n &= \sum_{j=0}^{n-1}\sum_{i=0}^{n}(-1)^{i+j}\delta_{n-1}^j\delta_n^i. \\
&= \sum_{0\leq i\leq j\leq n-1}(-1)^{i+j}\delta_{n-1}^j\delta_n^i + \sum_{0\leq j<i\leq n}(-1)^{i+j}\delta_{n-1}^j\delta_n^i \\
&= \sum_{0\leq i\leq j\leq n-1}(-1)^{i+j}\delta_{n-1}^i\delta_n^{j+1} + \sum_{0\leq j<i\leq n}(-1)^{i+j}\delta_{n-1}^j\delta_n^i \\
&= \sum_{0\leq j'<i'\leq n}(-1)^{i'+j'-1}\delta_{n-1}^{j'}\delta_n^{i'} + \sum_{0\leq j<i\leq n}(-1)^{i+j}\delta_{n-1}^j\delta_n^i \\
&= 0,
\end{aligned}$$

where in the second to last equality, we re-indexed the first sum using $i' = j+1$ and $j' = i$. Notice that the indexing of the final sum therefore changes from $0 \leq i \leq j \leq n-1$ to $0 \leq j' < i' \leq n$. This ensures that $j' = i$ still runs from 0 to $n-1$, always staying strictly below $i' = j+1$, which is equivalent to being less than or equal to j. The two sums clearly cancel, meaning that the bar complex is a complex. □

We now wish to show that $B(A)$ is not only a complex, it is an exact complex except for in degree 0.

Lemma 22. *$H_n(B(A)) = 0$ for all $n \neq 0$.*

Proof. The statement is equivalent to $B(A)$ being exact in all degrees except zero. To show exactness we insert A in degree -1 of $B(A)$. The resulting sequence looks like $B(A)$ except for at the tail, where we have

$$\cdots \xrightarrow{d_3} A^{\otimes 4} \xrightarrow{d_2} A^{\otimes 3} \xrightarrow{d_1} A^{\otimes 2} \xrightarrow{d_0} A \xrightarrow{0} 0,$$

where $d_0 : A \otimes A \longrightarrow A$ is the multiplication map, i.e. $d_0(a_0 \otimes a_1) = a_0a_1$, for $a_0, a_1 \in A$. If the above sequence is exact then $B(A)$ must be exact except for in degree 0.

We will now briefly forget the left A-module structure of the objects and morphisms in the bar complex, meaning we prove exactness while considering $B(A)$ as a complex of right A-modules. For any $n \in \mathbb{Z}$ we introduce a morphism of right A-modules $s_n : A^{\otimes n+2} \longrightarrow A^{\otimes n+3}$ by

$$s_n(a_0 \otimes \cdots \otimes a_{n+1}) = 1 \otimes a_0 \otimes \cdots \otimes a_{n+1}.$$

Notice that s_n is not left A-linear, but it is right A-linear. For $n < -1$, s_n is just the 0-map. This can be visualized as

$$\begin{tikzcd}
\cdots \arrow[r, "d_{n+3}"] & A^{\otimes n+4} \arrow[r, "d_{n+2}"] \arrow[d, "\mathrm{id}"] & A^{\otimes n+3} \arrow[r, "d_{n+1}"] \arrow[d, "\mathrm{id}"] \arrow[dl, "s_{n+2}"'] & A^{\otimes n+2} \arrow[r, "d_n"] \arrow[d, "\mathrm{id}"] \arrow[dl, "s_{n+1}"'] & A^{\otimes n+1} \arrow[r, "d_{n-1}"] \arrow[d, "\mathrm{id}"] \arrow[dl, "s_n"'] & \cdots \arrow[dl, "s_{n-1}"'] \\
\cdots \arrow[r, "d_{n+3}"] & A^{\otimes n+4} \arrow[r, "d_{n+2}"] & A^{\otimes n+3} \arrow[r, "d_{n+1}"] \arrow[dl, "s_{n+2}"'] & A^{\otimes n+2} \arrow[r, "d_n"] \arrow[dl, "s_{n+1}"'] & A^{\otimes n+1} \arrow[r, "d_{n-1}"] \arrow[dl, "s_n"'] & \cdots \arrow[dl, "s_{n-1}"']
\end{tikzcd}$$

We will now demonstrate that the identity map on the bar complex is null-homotopic, using the family of maps s_n. This would mean that the zero map is an isomorphism in homology. Consequently, all homology groups vanish and the complex would be exact. We need

$$\mathrm{id} = s_{n-1}d_n + d_{n+1}s_n,$$

for all $n \in \mathbb{Z}$. Let $\alpha = a_0 \otimes \cdots \otimes a_{n+1} \in A^{n+2}$, we compute

$$\begin{aligned}
(s_{n-1}d_n + d_{n+1}s_n)(\alpha) &= 1 \otimes d_n(\alpha) + d_{n+1}(1 \otimes \alpha) \\
&= 1 \otimes (\sum_{i=0}^{n}(-1)^i \delta_n^i(\alpha)) + (\sum_{i=0}^{n+1}(-1)^i \delta_{n+1}^i(1 \otimes \alpha)) \\
&= 1 \otimes (\sum_{i=0}^{n}(-1)^i \delta_n^i(\alpha)) + 1 \otimes (\sum_{i=0}^{n}(-1)^{i+1} \delta_n^i(\alpha))) + \alpha \\
&= \alpha.
\end{aligned}$$

In the penultimate equality, we take the first summand out of the second sum, and then we utilize the fact that δ_n^i does not affect the first tensor factor for $i \geq 1$. This allows us to pull the first tensor factor out of the sum. We then re-index the sum to make it clear that the sums cancel. Thus, the complex is exact, meaning $B(A)$ is exact outside of position 0. □

It is natural to ask what the only non-zero homology group of $B(A)$ is. We will see that this is clear from the computation of the above lemma.

Lemma 23. *$H_0(B(A)) = A$.*

Proof. The statement follows from the same argument as in Lemma 22. If inserting A in degree -1 of $B(A)$ makes the sequence exact, then it must necessarily be the cokernel of d_1, i.e. $H_0(B(A)) = \mathrm{coker}(d_1) = A$. □

By the above lemmas, the bar complex is a complex of A^e-modules which is exact in all degrees except 0 where the homology is A. This is almost everything that is required for $B(A)$ to be a projective resolution of A over A^{e}. The only missing piece is that $B(A)$ consists of projective objects.

Lemma 24. *The objects in the bar complex are projective A^e-modules.*

Proof. Firstly, observe that $A^{\otimes n+2} \cong A^{\mathrm{e}} \otimes_{\Bbbk} A^{\otimes n}$ as A^{e}-modules, by the assignment

$$(a_0 \otimes a_{n+1}) \otimes (a_1 \otimes \cdots \otimes a_n) \mapsto a_0 \otimes a_1 \otimes \cdots \otimes a_n \otimes a_{n+1},$$

for $a_0, \ldots, a_{n+1} \in A$. The A^{e}-module action on $A^{\mathrm{e}} \otimes_{\Bbbk} A^{\otimes n}$ is simply multiplication in the left tensor factor, A^{e}. The above assignment, is then A^{e}-linear and it is clearly bijective, so it is an isomorphism of A^{e}-modules as claimed.

In Example 12 we established that all free modules are projective. It therefore suffices to show that all $A^{\otimes n+2}$ are free as A^{e}-modules. Since $\Bbbk$ is a field, all modules

over $\Bbbk$ are free. In particular $A^{\otimes n}$ has a $\Bbbk$-basis. Let $\{\alpha_i | i \in I\}$, for an indexing set I, be such a $\Bbbk$-basis for $A^{\otimes n}$. We then have the following isomorphisms of A^{e}-modules

$$A^{\otimes n+2} \cong A^{\mathrm{e}} \otimes A^{\otimes n} \cong \bigoplus_{i \in I} A^{\mathrm{e}}(1 \otimes 1 \otimes \alpha_i).$$

Thus, $A^{\otimes n+2}$ is free as an A^{e}-module, and all the modules in the bar complex are projective over A^{e}. □

Combining the lemmas, it is clear that the bar complex is a projective resolution of A.

Theorem 25. *The bar complex is a projective resolution of A over A^e.*

Proof. This follows from Lemmas 21 to 24. □

Because of this the bar complex is sometimes called the bar resolution or standard resolution, when working over a field. Notice that if we were not working over a field, Lemma 24 would not hold and the bar complex might not be a projective resolution.

3.3 Definition of Hochschild cohomology

We are now ready to define Hochschild cohomology. Let M be an A^{e}-module. The Hochschild cohomology of A with coefficients in M is the cohomology of the complex obtained by applying $\mathrm{Hom}_{A^e}(-, M)$ to the bar resolution. Notice that $\mathrm{Hom}_{A^e}(-, M)$ is a contravariant functor, and therefore turns the bar resolution into a cochain complex.

We apply $\mathrm{Hom}_{A^e}(-, M)$ to the bar resolution and get

$$0 \longrightarrow \mathrm{Hom}_{A^e}(A \otimes A, M) \xrightarrow{d_1^*} \mathrm{Hom}_{A^e}(A^{\otimes 3}, M) \xrightarrow{d_2^*} \cdots, \tag{3.3.1}$$

where $d_n^* : \mathrm{Hom}_{A^e}(A^{\otimes n+1}, M) \longrightarrow \mathrm{Hom}_{A^e}(A^{\otimes n+2}, M)$ is induced by d_n, meaning that $d_n^*(f) = f d_n$ for $f \in \mathrm{Hom}_{A^e}(A^{\otimes n+1}, M)$.

Definition 26. The *n-th Hochschild cohomology* of a $\Bbbk$-algebra A, with coefficients in an A-bimodule M, is defined as the n-th cohomology of $\mathrm{Hom}_{A^e}(B(A), M)$, meaning the n-th cohomology of the complex (3.3.1), i.e.

$$HH^n(A, M) = \ker(d_{n+1}^*)/\mathrm{im}(d_n^*).$$

The *Hochschild cohomology* of A with coefficients in M is

$$HH^*(A, M) = \bigoplus_{n \geq 0} HH^n(A, M).$$

When computing the Hochschild cohomology of A with coefficients in itself, we abbreviate the notation to $HH^n(A) = HH^n(A, A)$ and $HH^*(A) = HH^*(A, A)$.

Remark 27. Notice that when $\Bbbk$ is a field, as in this thesis, the bar resolution is a projective resolution and therefore $HH^n(A, M) \cong \mathrm{Ext}^n_{A^e}(A, M)$. As a consequence of this, we can choose any projective resolution of A as an A^e-module and make computations using that resolution. This is because Ext does not depend on the choice of resolution. For a proof of this, and more on Ext, see [8].

3.4 Computations of Hochschild cohomology

The aim of this subsection is to do some computations. We start by building some intuition on what the Hochschild cohomology groups in the lower degrees correspond to. In the first two examples examples, let A be a $\Bbbk$-algebra and M an A-bimodule. We will use the notation of (3.3.1). We also compute the Hochschild cohomology groups for the two algebras $\Bbbk$ and $\Bbbk[x]/(x^2)$.

Example 28. Degree 0

We wish to compute $HH^0(A, M) = \ker(d_1^*)$, using the notation of the complex (3.3.1). Consider $f \in \mathrm{Hom}_{A^e}(A \otimes A, M)$. Let $f(1 \otimes 1) = m \in M$. Notice that, $f(a \otimes b) = af(1 \otimes 1)b = amb$, for any $a, b \in A$ due to A^e-linearity. Thus, f is completely determined by $f(1 \otimes 1)$.

Now assume that $f \in \ker(d_1^*)$, i.e. $fd_1 = 0$, so for any $a \in A$:

$$\begin{aligned} 0 &= d_1^*(f)(1 \otimes a \otimes 1) = f(d_1(1 \otimes a \otimes 1)) \\ &= f(a \otimes 1 - 1 \otimes a) = af(1 \otimes 1) - f(1 \otimes 1)a \\ &= am - ma. \end{aligned}$$

Thus, any f in the kernel, must satisfy $f(1 \otimes 1) = m$ such that $am = ma$ for any $a \in A$. Conversely, any element $m \in M$ such that $am = ma$ for all $a \in A$, determines a unique morphism $f_m \in \ker(d_1^*)$ by $f_m(1 \otimes 1) = m$. Hence, there exists an isomorphism of $\Bbbk$-modules $\ker(d_1^*) \cong \{m \in M \mid am = ma,\ \forall a \in A\}$ sending f to $f(1 \otimes 1)$, meaning that

$$HH^0(A, M) \cong \{m \in M \mid am = ma, \text{ for all } a \in A\}.$$

If $M = A$, this means that the 0-th Hochschild cohomology is simply the center of A,

$$HH^0(A) = Z(A).$$

The identification of the 0-th Hochschild cohomology group provides valuable intuition. We therefore aim to do the same for degree 1.

Example 29. Degree 1

Consider now $HH^1(A, M) = \ker(d_2^*)/\mathrm{im}(d_1^*)$, in (3.3.1). Let $g \in \ker(d_2^*)$, then for any $a, b \in A$, we have

$$\begin{aligned} 0 &= d_2^*(g)(1 \otimes a \otimes b \otimes 1) = g(d_2(1 \otimes a \otimes b \otimes 1)) \\ &= g(a \otimes b \otimes 1) - g(1 \otimes ab \otimes 1) + g(1 \otimes a \otimes b) \\ &= ag(1 \otimes b \otimes 1) - g(1 \otimes ab \otimes 1) + g(1 \otimes a \otimes 1)b. \end{aligned}$$

By abuse of notation, let $g(a)$ denote $g(1 \otimes a \otimes 1)$. Observe that all morphisms in the kernel of d_2^* fulfill $g(ab) = ag(b) + g(a)b$. This is exactly the definition of *derivations from A to M*, denoted by $\mathrm{Der}(A, M)$.

As obtained in the computations for degree 0, the morphisms in $\mathrm{Hom}_{A^e}(A \otimes A, M)$ are all of the form f_m, where $f_m(a \otimes b) = amb$, for some $m \in M$. Thus,

$$\mathrm{im}(d_1^*) = \{am - ma \mid a \in A, m \in M\}.$$

Functions on this form are called the *inner $\Bbbk$-derivations from A to M*, denoted by $\mathrm{InnDer}(A, M)$. We conclude that

$$HH^1(A, M) = \mathrm{Der}(A, M)/\mathrm{InnDer}(A, M),$$

which is often referred to as the *outer derivations* from A to M. If $M = A$, this is the outer derivations on A. If A is commutative, there are no inner derivations, and the Hochschild cohomology in degree 1 is just the derivations on A.

We now have a complete understanding of the first two Hochschild cohomology groups.

Remark 30. Although the computations are omitted in this thesis, it is an important result that $HH^2(A)$ categorizes the so-called *infinitesimal deformations* on A. For more on this see [12].

We move on to describe explicitly the Hochschild cohomology-groups of two basic algebras.

Example 31. $A = \Bbbk$

We compute a basic example. Consider $\Bbbk$ an algebra over itself. We can then construct the minimal projective resolution of $\Bbbk$ in ${}_{\Bbbk^e}\mathrm{Mod}$ as follows

$$\cdots \longrightarrow 0 \longrightarrow \Bbbk \longrightarrow 0 \longrightarrow \cdots$$

Let the one copy of $\Bbbk$ be in degree 0. Then the 0-th homology of the complex is $\Bbbk$, while in every other degree we get 0. Notice that $\Bbbk^{\mathrm{op}} = \Bbbk$, as $\Bbbk$ is commutative. Thus, $\Bbbk^e = \Bbbk \otimes_{\Bbbk} \Bbbk \cong \Bbbk$, so $\Bbbk$ is free as a $\Bbbk^e$-module. The above complex is therefore a projective resolution of $\Bbbk$ over $\Bbbk^e$.

We apply $\mathrm{Hom}_{\Bbbk^e}(-, M)$ for a $\Bbbk$-bimodule M. We get

$$\cdots \longrightarrow 0 \longrightarrow \mathrm{Hom}_{\Bbbk^e}(\Bbbk, M) \longrightarrow 0 \longrightarrow \cdots$$

Clearly $\mathrm{Hom}_{\Bbbk^e}(\Bbbk, M) \cong M$, as any morphism is uniquely determined by what $1 \in \Bbbk$ is mapped to. We conclude that

$$HH^n(\Bbbk, M) = \begin{cases} M, & n = 0 \\ 0, & \text{otherwise.} \end{cases}$$

Notice that since we require that the left and right $\Bbbk$-action on M agree, we get

$$\{m \in M \mid km = mk, \text{ for all } k \in \Bbbk\} = M,$$

so our conclusion agrees with the identification of $HH^0(\Bbbk, M)$ in Example 28. Moreover, there are no derivations from $\Bbbk$ to M, as for $d \in \mathrm{Der}(\Bbbk, M)$, evaluation of 1 yields

$$d(1) = d(1 \cdot 1) = 1d(1) + d(1)1 = 2d(1),$$

hence $d(1) = 0$ and since 1 is the generator of $\Bbbk$, $d = 0$. Thus, there are no non-zero derivations, as we would expect.

Now we consider a less trivial example, which we will revisit in Section 4.3.

Example 32. $A = \Bbbk[x]/(x^2)$

Let $A = \Bbbk[x]/(x^2)$. We compute $HH^n(A)$ for all $n \in \mathbb{Z}$. To do this, we define a variant of the bar resolution, namely the normalized bar resolution $\bar{B}(A)$. Let $\bar{A} = A/\Bbbk$, where $\Bbbk$ is regarded as the A-submodule $\Bbbk \cdot 1_A$. We define the normalized bar resolution as

$$\bar{B}(A) : \cdots \xrightarrow{d_3} A \otimes \bar{A}^{\otimes 2} \otimes A \xrightarrow{d_2} A \otimes \bar{A} \otimes A \xrightarrow{d_1} A^{\otimes 2} \xrightarrow{0} 0,$$

where the differentials are completely analogous to those of the bar resolution. The arguments for the normalized bar complex being a projective resolution of A over A^e are analogous as for the bar resolution. In our case $\bar{A}$ is generated by the single element x, i.e. $\bar{A} \cong \Bbbk\{x\} \cong \Bbbk$. We also observe that for $a_0, a_1 \in A$, we have

$$d_n(a_0 \otimes x \otimes \cdots \otimes x \otimes a_1) = a_0 x \otimes x \otimes \cdots \otimes x \otimes a_1 - 0 + 0 - \cdots + (-1)^n a_0 \otimes x \otimes \cdots \otimes x \otimes a_1 x,$$

as $x^2 = 0$ in A. Note that we write only x in the place of an arbitrary element, $kx \in \bar{A}$, for $k \in \Bbbk$. This is sufficient, as the tensor product is over $\Bbbk$ and the differentials are $\Bbbk$-linear.

We define A^{e}-module homomorphisms $\partial_n : A \otimes_{\Bbbk} A \longrightarrow A \otimes_{\Bbbk} A$, by sending

$$1 \otimes 1 \mapsto x \otimes 1 + (-1)^n 1 \otimes x$$

for $n \in \mathbb{N}$. It is easy to see that $\partial_{n-1}\partial_n = 0$ for all n, so these differentials make a complex. Notice that the diagram

$$\begin{tikzcd}
\cdots \arrow[r, "d_{n+1}"] & A \otimes_{\Bbbk} \bar{A}^{\otimes n} \otimes_{\Bbbk} A \arrow[r, "d_n"] \arrow[d, "\cong"] & A \otimes_{\Bbbk} \bar{A}^{\otimes n-1} \otimes_{\Bbbk} A \arrow[r, "dn-1"] \arrow[d, "\cong"] & \cdots \\
\cdots \arrow[r, "\partial_{n+1}"] & A \otimes_{\Bbbk} A \arrow[r, "\partial_n"] & A \otimes_{\Bbbk} A \arrow[r, "\partial_{n-1}"] & \cdots
\end{tikzcd}$$

commutes, where the isomorphism simply sends $1 \otimes x \otimes \cdots \otimes x \otimes 1 \mapsto 1 \otimes 1$. The complexes are therefore isomorphic. Applying $\mathrm{Hom}_{A^e}(-, A)$ to the lower sequence results in

$$\begin{tikzcd}
\cdots \arrow[r, "\partial^*_{n-1}"] & \mathrm{Hom}_{A^e}(A \otimes_{\Bbbk} A, A) \arrow[r, "\partial^*_n"] \arrow[d, "\cong"] & \mathrm{Hom}_{A^e}(A \otimes_{\Bbbk} A, A) \arrow[r, "\partial^*_{n+1}"] \arrow[d, "\cong"] & \cdots \\
\cdots \arrow[r, "\partial^*_{n-1}"] & A \arrow[r, "\partial^*_n"] & A \arrow[r, "\partial^*_{n+1}"] & \cdots
\end{tikzcd}$$

Notice that we have identified, $\mathrm{Hom}_{A^e}(A \otimes_{\Bbbk} A, A)$ with A, by simply sending f to $f(1 \otimes 1)$. This follows from the fact that $A \otimes_{\Bbbk} A \cong A^e$ as an A^e-module. We use the same name for the differentials $\partial^*_n : A \longrightarrow A$, and notice that $\partial^*_n(a) = xa + (-1)^n ax$ makes the diagram commute. As A is commutative, the differential is just zero for n odd, and multiplication by $2x$ for n even. Finally, we get

$$0 \longrightarrow A \xrightarrow{0} A \xrightarrow{\cdot 2x} A \xrightarrow{0} A \xrightarrow{\cdot 2x} \cdots .$$

If the characteristic of $\Bbbk$ is 2, then all the maps are 0, resulting in

$$HH^n(A) \cong \begin{cases} A, & n \geq 0 \\ 0, & n < 0 \end{cases}$$

If the characteristic of $\Bbbk$ is not 2, we get

$$HH^n(A) \cong \begin{cases} A, & n = 0 \\ (x), & n \geq 1 \text{ odd} \\ A/(x), & n > 1 \text{ even} \end{cases}$$

Notice that $(x) \cong \Bbbk \cong A/(x)$, so we can simplify to

$$HH^n(A) \cong \begin{cases} A, & n = 0 \\ \Bbbk, & n \geq 1 \end{cases}$$

Again we see that, as A is commutative, $Z(A) = A$, so our result agrees with Example 28.

Moreover, consider a $\Bbbk$-derivation $d \in \mathrm{Der}(A, A)$. By the same argument as in Example 31, $d(1) = 0$. It must also satisfy

$$0 = d(0) = d(x^2) = xd(x) + d(x)x = 2d(x).$$

Thus, $d(x)$ must be a $\Bbbk$-multiple of x, as otherwise $xd(x) \neq 0$. Hence, we get $\mathrm{Der}(A, A) \cong \Bbbk$, and as A is commutative, there are no inner derivations, so our computations agrees with Example 29.

4 The Bardzell resolution

As explained in the previous section, the Hochschild cohomology is invariant under the choice of projective resolution as long as we are working over a field. It is therefore very useful to become familiar with some alternative constructions of projective resolutions as candidates for computation. In this section we will construct the Bardzell resolution which can be defined for any monomial algebra. We, however, restrict our attention to the special case of quadratic monomial algebras, which results in an especially manageable resolution. Note that both gentle and almost gentle algebras are quadratic monomial [4]. The construction we present is the one used to describe the Hochschild cohomology for gentle algebras in [3], and the one that we will later use in order to investigate the broader class of almost gentle algebras.

4.1 Quadratic monomial algebras

If $\Bbbk$ is algebraically closed, then any finite $\Bbbk$-algebra is Morita equivalent to a path algebra modulo some admissible ideal, i.e. $\Bbbk Q/I$ for some quiver Q and ideal $I \subseteq \Bbbk Q$ [1]. We say that a finite dimensional algebra $A = \Bbbk Q/I$ is *monomial* if the ideal I is generated by monomials. We use the notation R for a minimal set of monomial generators of I and often refer to it as the set of relations. Let $\mathcal{B}$ be the set of paths in Q with no subpaths in R. $\mathcal{B}$ is then a $\Bbbk$-basis for A. If all paths in R are of length 2, we call the algebra *quadratic* monomial. It is exactly these quadratic monomial algebras which will be the subject of study in the remainder of the thesis.

Example 33. The algebra described in Example 18, $\Bbbk[x]/(x^n)$, is an example of a monomial algebra. If $n = 2$, the ideal would be generated by a single length 2 monomial, so the algebra would be quadratic monomial.

4.2 Construction of the Bardzell resolution

We are now ready to construct the Bardzell resolution. We follow the construction presented in [3]. Let Q be a quiver and $I = \Bbbk R \subseteq \Bbbk Q$ be an ideal generated by paths of length 2 in Q, making $A = \Bbbk Q/I$ a quadratic monomial algebra. We denote the vector space spanned by a set of paths P by $\Bbbk P$. For the paths of length zero, it is then $\Bbbk Q_0$. We consider $\Bbbk P$ as a $\Bbbk Q_0$-bimodule in the obvious way.

Let Γ_n be the set of paths in Q of length n which have no subpaths of length 2 that are not in R. Notice that $\Gamma_0 = Q_0$, $\Gamma_1 = Q_1$ and $\Gamma_2 = R$. We now consider $\Bbbk\Gamma_n$ as $\Bbbk Q_0$-bimodules.

Definition 34. The *Bardzell resolution* for a quadratic monomial algebra $A = \Bbbk Q/I$ is given as

$$\Gamma(A): \quad \cdots \xrightarrow{\Delta_{n+1}} A \otimes_{\Bbbk Q_0} \Bbbk\Gamma_n \otimes_{\Bbbk Q_0} A \xrightarrow{\Delta_n} \cdots \xrightarrow{\Delta_1} A \otimes_{\Bbbk Q_0} \Bbbk\Gamma_0 \otimes_{\Bbbk Q_0} A \longrightarrow 0,$$

with $\Gamma_n(A) = A \otimes_{\Bbbk Q_0} \Bbbk\Gamma_n \otimes_{\Bbbk Q_0} A$ in degree n. The differentials are defined as

$$\Delta_n(1 \otimes \alpha_1\alpha_2 \dots \alpha_n \otimes 1) = \alpha_1 \otimes \alpha_2 \dots \alpha_n \otimes 1 + (-1)^n 1 \otimes \alpha_1 \dots \alpha_{n-1} \otimes \alpha_n,$$

for $\alpha_i \in Q_1$ such that $\alpha_1\alpha_2 \dots \alpha_n$ is in Γ_n.

We will later see that this resolution can be very useful in computation. Already, one can notice that where the bar complex grows infinitely in the number of tensor factors, the Bardzell resolution stays at a manageable three factors. The differentials are also considerably easier to work with, as they only contain two summands. Before we can work with the Bardzell resolution, however, we must show that it is truly a projective resolution of A over A^{e}. To do this, we follow a similar pattern as for when the bar resolution was proven to be a projective resolution in Section 3.2.

Lemma 35. $\Gamma(A)$ *is a complex of* A^e*-modules.*

Proof. It is clear that the objects in the Bardzell resolution are A^{e}-modules, and that the differentials are A^{e}-linear. To observe this, simply recall that the module action can be defined on the left-most and right-most tensor factor, in which every object has a copy of A.

We must now confirm that $\Delta_{n-1}\Delta_n = 0$ for all $n \geq 2$. To do this, we make use of the A^{e} linearity of the differentials. For a path $\alpha = \alpha_1 \dots \alpha_n$ in Γ_n, we get

$$\begin{aligned}
\Delta_{n-1}\Delta_n(1 \otimes \alpha \otimes 1) &= \Delta_{n-1}(\alpha_1 \otimes \alpha_2 \dots \alpha_n \otimes 1 + (-1)^n 1 \otimes \alpha_1 \dots \alpha_{n-1} \otimes \alpha_n) \\
&= \alpha_1 \Delta_{n-1}(1 \otimes \alpha_2 \dots \alpha_n \otimes 1) + (-1)^n \Delta_{n-1}(1 \otimes \alpha_1 \dots \alpha_{n-1} \otimes 1)\alpha_n \\
&= 0 \otimes \alpha_3 \dots \alpha_n \otimes 1 + (-1)^{n-1}\alpha_1 \otimes \alpha_2 \dots \alpha_{n-1} \otimes \alpha_n \\
&\quad + (-1)^n(\alpha_1 \otimes \alpha_2 \dots \alpha_{n-1} \otimes \alpha_n + (-1)^{n-1} 1 \otimes \alpha_1 \dots \alpha_{n-2} \otimes 0 \\
&= 0.
\end{aligned}$$

Here, we observe that the two middle terms cancel and recall that $\alpha_i\alpha_{i+1} \in R$ for $0 \leq i < n$. Hence, the Bardzell resolution is a complex. □

The next step is to show that the A^{e}-modules in the Bardzell resolution are projective. To do this we recall the identification of projective modules stated in Section 2.3.

Lemma 36. *The objects of* $\Gamma(A)$ *are all projective as* A^e*-modules.*

Proof. To prove the above statement, first observe that

$$A \cong \bigoplus_{j \in Q_0} e_j A,$$

as a left $\Bbbk Q_0$-module through the $\Bbbk Q_0$-linear isomorphism given by

$$a \mapsto \bigoplus_{j \in Q_0} e_j a,$$

for any $a \in A$. Note that $\bigoplus_{j \in Q_0} e_j A$ is a left $\Bbbk Q_0$-module in the obvious way, but not a left A-module, only a right A-module. By analogous arguments the following isomorphism of right $\Bbbk Q_0$ modules also holds:

$$A \cong \bigoplus_{i \in Q_0} A e_i.$$

Furthermore, by definition, we have

$$\Bbbk \Gamma_n = \bigoplus_{\gamma \in \Gamma_n} \Bbbk \gamma,$$

for any n. Thus, we can rewrite the Bardzell objects as

$$A \otimes_{\Bbbk Q_0} \Bbbk \Gamma_n \otimes_{\Bbbk Q_0} A \cong (\bigoplus_{i \in Q_0} A e_i) \otimes_{\Bbbk Q_0} (\bigoplus_{\gamma \in \Gamma_n} \Bbbk \gamma) \otimes_{\Bbbk Q_0} (\bigoplus_{j \in Q_0} e_j A).$$

Distributing the direct sum yields

$$A \otimes_{\Bbbk Q_0} \Bbbk \Gamma_n \otimes_{\Bbbk Q_0} A \cong \bigoplus_{\gamma \in \Gamma_n} A e_{e_s(\gamma)} \otimes_{\Bbbk Q_0} \Bbbk \gamma \otimes_{\Bbbk Q_0} e_{t(\gamma)} A.$$

Note that we have identified many of the direct summands as zero, because the idempotents e_i and e_j do not match up with $s(\gamma)$ and $t(\gamma)$. This results in the zero tensor when the tensor product is over $\Bbbk Q_0$. Now, observe that because of this fact all idempotents match up in each direct summands. Therefore, the tensor products can be seen as to be taken over $\Bbbk$, with a copy of $\Bbbk$ in the middle tensor factor. Hence, for any $\gamma \in \Gamma_n$ we get

$$A e_{e_s(\gamma)} \otimes_{\Bbbk Q_0} \Bbbk \gamma \otimes_{\Bbbk Q_0} e_{t(\gamma)} A \cong A e_{e_s(\gamma)} \otimes_{\Bbbk} \Bbbk \otimes_{\Bbbk} e_{t(\gamma)} A \cong A e_{e_s(\gamma)} \otimes_{\Bbbk} e_{t(\gamma)} A.$$

We will now argue that every $A e_{e_s(\gamma)} \otimes_{\Bbbk} e_{t(\gamma)} A$ is projective. We apply the same isomorphism as before and notice

$$A \otimes_{\Bbbk} A \cong (\bigoplus_{i \in Q_0} A e_i) \otimes_{\Bbbk} (\bigoplus_{j \in Q_0} e_j A) = \bigoplus_{i,j \in Q_0} A e_i \otimes_{\Bbbk} e_j A.$$

Thus, for every $\gamma \in \Gamma_n$, $A e_{e_s(\gamma)} \otimes_{\Bbbk} e_{t(\gamma)} A$ is a direct summand of $A \otimes_{\Bbbk} A$. As we have remarked previously, we know $A \otimes_{\Bbbk} A \cong A^{\mathrm{e}}$ as an A^{e}-module. Importantly this means $A \otimes_{\Bbbk} A$ is free as an A^{e}-module. As stated in Section 2.3, a direct summand of a free module is projective. Thus, for any $\gamma \in \Gamma_n$, $A e_{e_s(\gamma)} \otimes_{\Bbbk} e_{t(\gamma)} A$ is projective and, since a direct sum of projectives is projective, so is $A \otimes_{\Bbbk Q_0} \Bbbk \Gamma_n \otimes_{\Bbbk Q_0} A$ for all n.

□

We have concluded that the Bardzell resolution is a complex of projective A^{e}-modules. It remains to show that it is exact in all degrees except 0, where the homology should be A.

Lemma 37. *$H_n(\Gamma(A)) = 0$ for all $n \neq 0$ and $H_0(\Gamma(A)) = A$.*

Proof. To prove the above statement, we add the augmentation $f : A \otimes \Gamma_0 \otimes A \longrightarrow A$ given by sending $1 \otimes e \otimes 1 \mapsto e$ for $e \in \Gamma_0 = Q_0$, which is clearly surjective. The resulting sequence is

$$\cdots \xrightarrow{\Delta_2} A \otimes_{\Bbbk Q_0} \Gamma_1 \otimes_{\Bbbk Q_0} A \xrightarrow{\Delta_1} A \otimes_{\Bbbk Q_0} \Gamma_0 \otimes_{\Bbbk Q_0} A \xrightarrow{f} A \longrightarrow 0.$$

Observe that for any $\alpha \in \Gamma_1 = Q_1$, we have

$$f\Delta_1(1 \otimes \alpha \otimes 1) = \alpha f(1 \otimes e_{t(\alpha)} \otimes 1) - f(1 \otimes e_{s(\alpha)} \otimes 1)\alpha = \alpha e_{t(\alpha)} - e_{s(\alpha)}\alpha = 0,$$

so the sequence is in fact a complex. If this complex is exact, then the statement in the lemma holds.

To show exactness, we will use a contracting homotopy introduced by Sköldberg in 2008 [11]. Similarly to in the proof of Lemma 22, we forget the left module structure. The contracting homotopy is, therefore, a family of morphisms s_n in the category ${}_{\Bbbk Q_0}\mathrm{Mod}_A$ for $n \in \mathbb{N}_0$. We let $s_n : A \otimes \Gamma_n \otimes A \longrightarrow A \otimes \Gamma_{n+1} \otimes A$ be given by

$$s_n(p \otimes \alpha \otimes e) = \sum_{\substack{q_1, q_3 \in \mathcal{B}, \\ q_2 \in \Gamma_{n+1}, \\ q_1 q_2 q_3 = p\alpha}} q_1 \otimes q_2 \otimes q_3,$$

for the basis element $p \otimes \alpha \otimes e$, where $p \in \mathcal{B}$, the $\Bbbk$-basis of A, $e \in Q_0$ and $\alpha = \alpha_1 \dots \alpha_n \in \Gamma_n$. As s_n is right A-linear, this is defined on the entire basis of $A \otimes \Bbbk\Gamma_n \otimes A$. Throughout this proof, we will write e for an element in Q_0 and always mean the vertex in Q_0 such that the necessary compositions make sense. We also consider the diagonal map $s_{-1} : A \longrightarrow A \otimes \Bbbk\Gamma_0 \otimes A$ defined by

$$s_{-1}(p) = e \otimes e \otimes p.$$

Putting this together, we get the following diagram:

$$\begin{tikzcd}
\cdots \arrow[r, "\Delta_3"] & A \otimes_{\Bbbk Q_0} \Gamma_2 \otimes_{\Bbbk Q_0} A \arrow[r, "\Delta_2"] \arrow[dl, "s_2"'] \arrow[d, "\mathrm{id}"] & A \otimes_{\Bbbk Q_0} \Gamma_1 \otimes_{\Bbbk Q_0} A \arrow[r, "\Delta_1"] \arrow[dl, "s_1"'] \arrow[d, "\mathrm{id}"] & A \otimes_{\Bbbk Q_0} \Gamma_0 \otimes_{\Bbbk Q_0} A \arrow[r, "f"] \arrow[dl, "s_0"'] \arrow[d, "\mathrm{id}"] & A \arrow[r] \arrow[dl, "s_{-1}"'] \arrow[d, "\mathrm{id}"] & 0 \arrow[dl, "0"'] \\
\cdots \arrow[r, "\Delta_3"] & A \otimes_{\Bbbk Q_0} \Gamma_2 \otimes_{\Bbbk Q_0} A \arrow[r, "\Delta_2"] & A \otimes_{\Bbbk Q_0} \Gamma_1 \otimes_{\Bbbk Q_0} A \arrow[r, "\Delta_1"] & A \otimes_{\Bbbk Q_0} \Gamma_0 \otimes_{\Bbbk Q_0} A \arrow[r, "f"] & A \arrow[r] & 0
\end{tikzcd}$$

Let us study the diagonal maps s_n. We start with the case where $n > 0$. Note that we then get

$$s_n(p \otimes \alpha_1 \dots \alpha_n \otimes e) = \begin{cases} p' \otimes \alpha_0 \dots \alpha_n \otimes e, & \text{if } p = p'\alpha_0 \text{ such that } \alpha_0\alpha_1 \in R \\ 0, & \text{otherwise.} \end{cases} \tag{4.2.1}$$

This is true as the listed element is necessarily the only element that satisfies the requirements of the sum in the definition of s_n. To justify this, notice that the middle tensor factor, which should increase in length by one, must take iteratively from the last arrows in p for the composition of the tensor factors to remain the same. However, if it was possible to take two arrows of p, then the two last arrows of p would make up an element of R, which would mean p is 0 in A. Moreover, if

the last arrow in p does not compose with α_1 into something in R, there is no way of making a suitable element of Γ_{n+1} while keeping the composition. Now assume there exists such a suitable decomposition, $p = p'\alpha_0$ of the first tensor factor. We compute the compositions $\Delta_{n+1}s_n$ and $s_{n-1}\Delta_n$ for $n > 1$ and get

$$\begin{aligned}\Delta_{n+1}s_n(p \otimes \alpha_1 \dots \alpha_n \otimes e) &= \Delta_{n+1}(p' \otimes \alpha_0\alpha_1 \dots \alpha_n \otimes e)\\ &= p \otimes \alpha_1 \dots \alpha_n \otimes e + (-1)^{n+1}p' \otimes \alpha_0\alpha_1 \dots \alpha_{n-1} \otimes \alpha_n,\end{aligned}$$

$$\begin{aligned}s_{n-1}\Delta_n(p \otimes \alpha_1 \dots \alpha_n \otimes e) &= s_{n-1}(p \otimes \alpha_2 \dots \alpha_n \otimes e + (-1)^n p \otimes \alpha_1 \dots \alpha_{n-1} \otimes \alpha_n)\\ &= 0 + (-1)^n p' \otimes \alpha_0\alpha_1 \dots \alpha_{n-1} \otimes \alpha_n.\end{aligned}$$

Clearly this results in $\Delta_{n+1}s_n + s_{n-1}\Delta_n = \mathrm{id}$, building towards our goal of showing that the identity is in fact null-homotopic. Now we look at the case where no appropriate decomposition of p exists, meaning that s_n will return 0. We get

$$\begin{aligned}(\Delta_{n+1}s_n + s_{n-1}\Delta_n)(p \otimes \alpha_1 \dots \alpha_n \otimes e) &= 0 + s_{n-1}\big(p\alpha_1 \otimes \alpha_2 \dots \alpha_n \otimes e)\\ &\quad + (-1)^n s_{n-1}(p \otimes \alpha_1 \dots \alpha_{n-1} \otimes \alpha_n\big)\\ &= p \otimes \alpha_1 \dots \alpha_n \otimes e + 0.\end{aligned}$$

Here we have used $\alpha_i\alpha_{i+1} = 0$ in A, for $0 \leq i \leq n-1$. Hence, the equality we desire holds for $n > 1$. We must now show it for degree $1, 0$ and -1. Let the path p be composed of l arrows in Q_1 such that $p = p_1 \dots p_l$. For s_0 , notice that since $\Gamma_1 = Q_1$ we get

$$s_0(p \otimes e \otimes e) = e \otimes p_1 \otimes p_2 \dots p_l + (\sum_{i=2}^{l-1} p_1 \dots p_{i-1} \otimes p_i \otimes p_{i+1} \dots p_l) + p_1 \dots p_{l-1} \otimes p_l \otimes e.$$

Now we wish to tackle degree 1. We aim to show that $\Delta_2 s_1 + s_0\Delta_1 = \mathrm{id}$. To do this, we must again consider the two cases in Equation (4.2.1). We consider p as above. First, we let $p \otimes \alpha_1 \otimes e \in A \otimes \Bbbk Q_1 \otimes A$ such that $p_l\alpha_1 \in R$. Now applying $\Delta_2 s_1$ and $s_0\Delta_1$ gives us

$$\begin{aligned}\Delta_2 s_1(p \otimes \alpha_1 \otimes e) &= \Delta_2(p_1 \dots p_{l-1} \otimes p_l\alpha_1 \otimes e)\\ &= p \otimes \alpha_1 \otimes e + p_1 \dots p_{l-1} \otimes p_l \otimes \alpha_1,\end{aligned}$$

$$\begin{aligned}s_0\Delta_1(p \otimes \alpha_1 \otimes e) &= s_0(0 - p \otimes e \otimes \alpha_1) = -s_0(p \otimes e \otimes e)\alpha_1\\ &= -\big(e \otimes p_1 \otimes p_2 \dots p_{l-1}p_l\alpha_1 + \sum_{i=2}^{l-1} p_1 \dots p_{i-1} \otimes p_i \otimes p_{i+1} \dots p_{l-1}p_l\alpha_1\\ &\quad + p_1 \dots p_{l-1} \otimes p_l \otimes \alpha_1\big)\\ &= -(0 + 0 + p_1 \dots p_{l-1} \otimes p_l \otimes \alpha_1).\end{aligned}$$

Again we see that the sum is the identity. In the second case, where the $p_l\alpha_1 \notin R$, $\Delta_2 s_1$ gives 0, so we get

$$\begin{aligned}(\Delta_2 s_1 + s_0\Delta_1)(p \otimes \alpha_1 \otimes e) &= 0 + s_0(p\alpha_1 \otimes e \otimes e - p \otimes e \otimes \alpha_1)\\ &= s_0(p\alpha_1 \otimes e \otimes e) - s_0(p \otimes e \otimes e)\alpha_1\\ &= p \otimes \alpha_1 \otimes e + s_0(p \otimes e \otimes e)\alpha_1 - s_0(p \otimes e \otimes e)\alpha_1\\ &= p \otimes \alpha_1 \otimes e.\end{aligned}$$

So far we have shown the desired equality holds for $n > 0$. We test for $n = 0$, which yields

$$\begin{aligned}
\Delta_1 s_0\big(p \otimes e \otimes e\big) &= \Delta_1\big(e \otimes p_1 \otimes p_2 \dots p_l + (\sum_{i=2}^{l-1} p_1 \dots p_{i-1} \otimes p_i \otimes p_{i+1} \dots p_l) \\
&\quad + p_1 \dots p_{l-1} \otimes p_l \otimes e\big) \\
&= (p_1 \otimes e \otimes p_2 \dots p_l - e \otimes e \otimes p) \\
&\quad + (p_1 p_2 \otimes e \otimes p_3 \dots p_l - p_1 \otimes e \otimes p_2 \dots p_l) + \cdots \\
&\quad \cdots + (p_1 \dots p_{l-1} \otimes e \otimes p_l - p_1 \dots p_{l-2} \otimes e \otimes p_{l-1} p_l) \\
&\quad + (p \otimes e \otimes e - p_1 \dots p_{l-1} \otimes e \otimes p_l) \\
&= p \otimes e \otimes e - e \otimes e \otimes p,
\end{aligned}$$

$$s_{-1} f(p \otimes e \otimes e) = s_{-1}(p) = e \otimes e \otimes p.$$

For the last equality of the first calculation we observed that each term appears twice, with opposite signs, except for the two remaining cases. Again the sum of the two compositions is the identity. Finally, it is clear that $f s_{-1} = \mathrm{id}$. Hence, the complex is exact, so the statement of the lemma holds. □

Combining the above lemmas we reach the desired conclusion.

Theorem 38. *The Bardzell resolution is a projective resolution of A over A^e.*

Proof. The statement follows from Lemmas 35 to 37. □

We now have an alternative projective resolution of A as an A^{e}-module, in the case where A is quadratic monomial. Given an A^{e}-module M, we could apply $\mathrm{Hom}_{A^{\mathrm{e}}}(-, M)$ to the Bardzell resolution. Computing cohomology will then result in the same as computing the Hochschild cohomology through the bar resolution. Using this, we will now do some computation.

4.3 Working with the Bardzell resolution

We finish the thesis by aiming to convince the reader that the Bardzell resolution defined in Section 4.2 is useful. This we will do by presenting a few examples. First we consider $A = \Bbbk[x]/(x^2)$, and observe how closely the computations coincide with the normalized bar resolution. We further construct two small quivers, give them relations, and study the algebras that arise. Importantly, we illustrate how the Bardzell resolution stops where the bar resolution does not, and use the convenient structure of the resolution to compute their Hochschild cohomology.

Example 39. $A = \Bbbk[x]/(x^2)$

We revisit $A = \Bbbk[x]/(x^2)$, as seen in Example 32. We will now compute $HH^n(A)$, using the Bardzell resolution. Recall that A is isomorphic to a quadratic monomial

algebra $\Bbbk Q/I$, with Q being the quiver with one vertex and one looping arrow, which has been redrawn below.

We define $I = \Bbbk\{x^2\}$, where x represents the arrow. Thus, the set of relations generating I is simply $R = \{x^2\}$. We will use the notation of Definition 34. First we note that

$$\begin{aligned}
&\Gamma_0 = Q_0 = \{e\} \\
&\Gamma_1 = Q_1 = \{x\} \\
&\Gamma_2 = R = \{x^2\} \\
&\Gamma_3 = \{x^3\} \\
&\cdots,
\end{aligned}$$

where e is the path of length 0 at the one vertex. In general we have $\Gamma_n = \{x^n\}$, meaning that $\Bbbk\Gamma_n \cong \Bbbk$ for all natural numbers n, through the isomorphism which sends $kx^n \longrightarrow k$, for $k \in \Bbbk$. Particularly, $\Bbbk Q_0 \cong \Bbbk$, so the tensor products in the Bardzell resolution can be seen to be taken over $\Bbbk$.

For all positive integers n, let $\partial_n : A \otimes_\Bbbk A \longrightarrow A \otimes_\Bbbk A$ be the A^{e}-linear morphism defined by sending $1 \otimes 1 \mapsto x \otimes 1 + (-1)^n 1 \otimes x$. Observe that $\partial_{n-1}\partial_n = 0$, so these differentials make a complex. We notice that

$$\begin{tikzcd}
\cdots \arrow[r, "\Delta_{n+1}"] & \Bbbk[x]/(x^2) \otimes_\Bbbk \Bbbk\Gamma_n \otimes_\Bbbk \Bbbk[x]/(x^2) \arrow[r, "\Delta_n"] \arrow[d, "\cong"] & \Bbbk[x]/(x^2) \otimes_\Bbbk \Bbbk\Gamma_{n-1} \otimes_\Bbbk \Bbbk[x]/(x^2) \arrow[r, "\Delta_{n-1}"] \arrow[d, "\cong"] & \cdots \\
\cdots \arrow[r, "\partial_{n+1}"] & \Bbbk[x]/(x^2) \otimes_\Bbbk \Bbbk[x]/(x^2) \arrow[r, "\partial_n"] & \Bbbk[x]/(x^2) \otimes_\Bbbk \Bbbk[x]/(x^2) \arrow[r, "\partial_{n-1}"] & \cdots
\end{tikzcd}$$

commutes, where the isomorphism simply sends $1 \otimes x^n \otimes 1 \longrightarrow 1 \otimes 1$. Thus, the complexes are isomorphic. The resulting projective resolution is the same as the one in Example 32. We therefore get the same result when computing Hochschild cohomology, namely

$$HH^n(A) = \begin{cases} A, & n = 0 \\ (x) \cong \Bbbk, & n \text{ odd} \\ A/(x) \cong \Bbbk, & n \text{ even} \end{cases}$$

in the case where the characteristic of $\Bbbk$ is different from 2.

It is worth remarking that, while we will always get the same result regardless of the choice of projective resolution, the calculations will in most cases not coincide as closely as this. In this specific case, for example, we had $\Bbbk Q_0 \cong \Bbbk$, which will not always be the case, so in general the tensor products will not be taken over the same ring in the two resolutions.

To illustrate how one can work directly with the Bardzell resolution, we compute the following two examples. Notice how the tensor product being over $\Bbbk Q_0$, along with A^{e}-linearity make the computations easier.

Example 40. Consider the quiver $1 \xrightarrow{\alpha} 2$ and denote it by Q. Let $A = \Bbbk Q$. Notice that A quadratic monomial, as the requirements on $I = 0$ are trivially satisfied. We identify Γ_n for all n as

$$\begin{aligned}
&\Gamma_0 = Q_0 = \{1, 2\},\\
&\Gamma_1 = Q_1 = \{\alpha\},\\
&\Gamma_2 = 0,\\
&\Gamma_3 = 0,\\
&\cdots .
\end{aligned}$$

We can see that our Bardzell resolution will be quite short, as $\Bbbk\Gamma_n = 0$ for $n \geq 2$, meaning that we simply get

$$0 \longrightarrow A \otimes_{\Bbbk Q_0} \Bbbk\{\alpha\} \otimes_{\Bbbk Q_0} A \xrightarrow{\Delta_1} A \otimes_{\Bbbk Q_0} \Bbbk Q_0 \otimes_{\Bbbk Q_0} A \longrightarrow 0.$$

Applying $\mathrm{Hom}_{A^e}(-, A)$ yields

$$0 \longrightarrow \mathrm{Hom}_{A^e}(A \otimes_{\Bbbk Q_0} \Bbbk Q_0 \otimes_{\Bbbk Q_0} A, A) \xrightarrow{\Delta_1^*} \mathrm{Hom}_{A^e}(A \otimes_{\Bbbk Q_0} \Bbbk\{\alpha\} \otimes_{\Bbbk Q_0} A, A) \longrightarrow 0.$$

Recall that in the proof of Lemma 36, we derived the following isomorphism:

$$A \otimes_{\Bbbk Q_0} \Bbbk\Gamma_n \otimes_{\Bbbk Q_0} A \cong \bigoplus_{\gamma \in \Gamma_n} Ae_{e_s(\gamma)} \otimes_{\Bbbk} e_{t(\gamma)}A.$$

Thus, we get

$$A \otimes_{\Bbbk Q_0} \Bbbk Q_0 \otimes_{\Bbbk Q_0} A \cong (Ae_1 \otimes_{\Bbbk} e_1 A) \oplus (Ae_2 \otimes_{\Bbbk} e_2)A$$

Applying this to the Hom-group, we obtain

$$\mathrm{Hom}_{A^e}(A \otimes_{\Bbbk Q_0} \Bbbk Q_0 \otimes_{\Bbbk Q_0} A, A) \cong \mathrm{Hom}_{A^e}(Ae_1 \otimes_{\Bbbk} e_1 A, A) \oplus \mathrm{Hom}_{A^e}(Ae_2 \otimes_{\Bbbk} e_2 A, A).$$

Let $f \in \mathrm{Hom}_{A^e}(Ae_i \otimes_{\Bbbk} e_i A, A)$, for $i \in 1, 2$. Since f is A-bilinear, for any $a_1 e_i \in Ae_i$ and $e_i a_2 \in e_i A$ we have

$$f(a_1 e_i \otimes e_i a_2) = a_1 f(e_i \otimes e_i) a_2,$$

and f is therefore determined by what $e_i \otimes e_i$ is sent to. Linearity forces

$$e_i f(e_i \otimes e_i) = f(e_i(e_i \otimes e_i)) = f((e_i \otimes e_i)e_i)) = f(e_i \otimes e_i)e_i.$$

Hence, $f(e_i \otimes e_i) = ce_i$ for some $c \in \Bbbk$. Conversely, for any $c \in \Bbbk$, ce_i determines a unique morphism by sending $(e_i \otimes e_i) \mapsto ce_i$. It follows that

$$\mathrm{Hom}_{A^e}(Ae_i \otimes_{\Bbbk} e_i A, A) \cong \Bbbk e_i \cong \Bbbk.$$

We move to $\mathrm{Hom}_{A^e}(A \otimes_{\Bbbk Q_0} \Bbbk\{\alpha\} \otimes_{\Bbbk Q_0} A, A)$. Again, using the isomorphism from Lemma 36 yields

$$A \otimes_{\Bbbk Q_0} \Bbbk\{\alpha\} \otimes_{\Bbbk Q_0} A \cong Ae_{s(\alpha)} \otimes_{\Bbbk} e_{t(\alpha)}A = Ae_1 \otimes_{\Bbbk} e_2 A$$

We identify the Hom-group in the same way as above. For $g \in \mathrm{Hom}_{A^e}(Ae_1 \otimes_{\Bbbk} e_2 A, A)$ we have

$$e_1 g(e_1 \otimes e_2) = g(e_1(e_1 \otimes e_2)) = g((e_1 \otimes e_2)e_2)) = g(e_1 \otimes e_2)e_2,$$

so $g(e_1 \otimes e_2)$ must be a $\Bbbk$-multiple of α, meaning $\mathrm{Hom}_{A^e}(A \otimes_{\Bbbk Q_0} \Bbbk\{\alpha\} \otimes_{\Bbbk Q_0} A, A)$ has $\Bbbk$-dimension 1. Thus, Δ_1^* sends a 2-dimensional space to a 1-dimensional space. In particular, this means that if Δ_1^* is not 0 it is surjective, since it is $\Bbbk$-linear. We show this by constructing a morphism, such that Δ_1^* sends it to something other than the 0-morphism.

Let $h \in \mathrm{Hom}_{A^e}(A \otimes_{\Bbbk Q_0} \Bbbk Q_0 \otimes_{\Bbbk Q_0} A, A)$, sending $1 \otimes (e_1 + e_2) \otimes 1 \mapsto e_2 - e_1$. We consider $\Delta_1^*(h) = h\Delta_1 \in \mathrm{Hom}_{A^e}(A \otimes_{\Bbbk Q_0} \Bbbk\{\alpha\} \otimes_{\Bbbk Q_0} A, A)$, it is non-zero because

$$h\Delta_1(1 \otimes \alpha \otimes 1) = h(\alpha \otimes e_2 \otimes 1 - 1 \otimes e_1 \otimes \alpha) = \alpha h(1 \otimes e_2 \otimes 1) - h(1 \otimes e_1 \otimes 1)\alpha = \alpha + \alpha \neq 0.$$

Thus, $\Delta_1^* \neq 0$ and it must be surjective. This, in turn, means

$$HH^1(A) = \ker(0)/\mathrm{im}(\Delta_1^*) = 0.$$

Since the image of Δ_1^* is $\Bbbk$ its kernel must also be isomorphic to $\Bbbk$ as it maps a 2-dimensional space to a 1-dimensional space. Therefore, we get

$$HH^0(A) = \ker(\Delta_1^*) = \Bbbk.$$

Notice that $Z(A) = \Bbbk\{e_1 + e_2\} \cong \Bbbk$, meaning our computation agree with the result from Section 3.4, that $HH^0(A) \cong Z(A)$.

Although the above example is of a very simple quiver, the methods used in the computations apply more generally when working with the Bardzell resolution. In the following lemma, we generalize one observation we made in the computations.

Lemma 41. *Let Q be a quiver, and A its path algebra. Then, as $\Bbbk$-modules $Hom_{A^e}(Ae_i \otimes_{\Bbbk} e_j A, A) \cong e_i A e_j$, for any e_i and $e_j \in Q_0$.*

Proof. Any morphism $f \in \mathrm{Hom}_{A^e}(Ae_i \otimes_{\Bbbk} e_j A, A)$ satisfies

$$f(a \otimes b) = af(e_i \otimes e_j)b,$$

for any a and $b \in A$. Thus, the morphism is determined by $f(e_i \otimes e_j)$. Observe that $f(e_i \otimes e_j) \in e_i A e_j$, as

$$e_i f(e_i \otimes e_j) e_j = f(e_i \otimes e_j).$$

In addition, every element in $e_i A e_j$ determines a unique morphism. Therefore, the assignment $f \mapsto f(e_i \otimes e_j)$ is a $\Bbbk$-linear bijection, and the statement of the lemma holds. □

Using this lemma we can compute the Hochschild cohomology for a more complex example.

Example 42. Consider one of the examples from Section 2.5, namely:

$$\begin{tikzcd}
 & & 3 \arrow[dr, "c"] & & \\
1 \arrow[r, "a"] & 2 \arrow[ur, "b"] \arrow[dr, "d"] \arrow[urr, dashed, bend left] & & 5 \arrow[r, "f"] & 6 \\
 & & 4 \arrow[ur, "e"] & &
\end{tikzcd}$$

Notice that we have added the following three relations, $R = \{ab, bc, cf\}$. We denote the quiver by Q and let $I = (R) \subset \Bbbk Q$. We consider the algebra $\Bbbk Q/I$, and observe that it is quadratic monomial. First, we identify the Γ-sets, we get

$$\begin{aligned}
&\Gamma_0 = Q_0 = \{1, 2, 3, 4, 5, 6\},\\
&\Gamma_1 = Q_1 = \{a, b, c, d, e, f\},\\
&\Gamma_2 = R = \{ab, bc, cf\},\\
&\Gamma_3 = \{abc, bcf\},\\
&\Gamma_4 = \{abcf\},\\
&\Gamma_5 = 0,\\
&\dots
\end{aligned}$$

giving us a manageable projective resolution of length 4. It is worth noting that for $A = \Bbbk Q/I$, $A^{\otimes n}$ is not 0 for any n, so the bar resolution would be infinite. This would result in significantly more cumbersome calculations.

We write down $\mathrm{Hom}_{A^e}(-, A)$ applied to the resolution. Recall that $\Gamma_n(A) = A \otimes_{\Bbbk Q_0} \Bbbk\Gamma_n \otimes_{\Bbbk Q_0} A$. This yields

$$\mathrm{Hom}_{A^e}(\Gamma_0(A), A) \xrightarrow{\Delta_1^*} \mathrm{Hom}_{A^e}(\Gamma_1(A), A) \xrightarrow{\Delta_2^*} \mathrm{Hom}_{A^e}(\Gamma_2(A), A) \xrightarrow{\Delta_3^*} \bullet$$

$$\bullet \xrightarrow{\Delta_3^*} (\mathrm{Hom}_{A^e}(\Gamma_3(A), A) \xrightarrow{\Delta_4^*} \mathrm{Hom}_{A^e}(\Gamma_4(A), A).$$

Applying the isomorphism from Lemma 36 and Lemma 41, we get

$$A \otimes_{\Bbbk Q_0} \Bbbk\Gamma_0 \otimes_{\Bbbk Q_0} A \cong \bigoplus_{i \in Q_0} Ae_i \otimes_{\Bbbk} e_i A \implies \mathrm{Hom}_{A^e}(A \otimes_{\Bbbk Q_0} \Bbbk\Gamma_0 \otimes_{\Bbbk Q_0} A, A) \cong \Bbbk^6,$$

$$A \otimes_{\Bbbk Q_0} \Bbbk\Gamma_1 \otimes_{\Bbbk Q_0} A \cong \bigoplus_{\alpha \in Q_1} Ae_{s(\alpha)} \otimes_{\Bbbk} e_{t(\alpha)} A \implies \mathrm{Hom}_{A^e}(A \otimes_{\Bbbk Q_1} \Bbbk\Gamma_1 \otimes_{\Bbbk Q_0} A, A) \cong \Bbbk^6,$$

$$\begin{gathered}
A \otimes_{\Bbbk Q_1} \Bbbk\Gamma_2 \otimes_{\Bbbk Q_0} A \cong Ae_1 \otimes_{\Bbbk} e_3 A \oplus Ae_2 \otimes_{\Bbbk} e_5 A \oplus Ae_3 \otimes_{\Bbbk} e_6 A\\
\implies \mathrm{Hom}_{A^e}(A \otimes_{\Bbbk Q_0} \Bbbk\Gamma_2 \otimes_{\Bbbk Q_0} A, A) \cong 0 \oplus \Bbbk \oplus 0 = \Bbbk,\\
A \otimes_{\Bbbk Q_1} \Bbbk\Gamma_3 \otimes_{\Bbbk Q_0} A \cong Ae_1 \otimes_{\Bbbk} e_5 A \oplus Ae_2 \otimes_{\Bbbk} e_6 A\\
\implies \mathrm{Hom}_{A^e}(A \otimes_{\Bbbk Q_0} \Bbbk\Gamma_3 \otimes_{\Bbbk Q_0} A, A) \cong \Bbbk \oplus \Bbbk,\\
A \otimes_{\Bbbk Q_1} \Bbbk\Gamma_4 \otimes_{\Bbbk Q_0} A \cong Ae_1 \otimes_{\Bbbk} e_6 A \implies \mathrm{Hom}_{A^e}(A \otimes_{\Bbbk Q_0} \Bbbk\Gamma_4 \otimes_{\Bbbk Q_0} A, A) \cong \Bbbk.
\end{gathered}$$

The final isomorphism in each of the above results are obtained using Lemma 41 and noting the relations. E.g. $\mathrm{Hom}_{A^e}(Ae_1 \otimes_{\Bbbk} e_3A, A) = 0$ as $e_1Ae_3 = \Bbbk\{ab\} \subseteq I$. However, $\mathrm{Hom}_{A^e}(Ae_2 \otimes_{\Bbbk} e_5A, A) \cong \Bbbk$ as $e_2Ae_5 = \Bbbk\{bc, de\} \cong \Bbbk$, and bc is a relation, but de is not.

Thus, $\mathrm{Hom}_{A^e}(-, A)$ applied to our Bardzell resolution can be considered as

$$0 \longrightarrow \Bbbk^6 \xrightarrow{\Delta_1^*} \Bbbk^6 \xrightarrow{\Delta_2^*} \Bbbk \xrightarrow{\Delta_3^*} \Bbbk^2 \xrightarrow{\Delta_4^*} \Bbbk \longrightarrow 0.$$

Now we use the fact that we know $HH^0(A) \cong Z(A)$. Observe that

$$Z(A) = \Bbbk \cdot 1_A = \Bbbk\{e_1 + e_2 + e_3 + e_4 + e_5 + e_6\},$$

as any other element is not central. This means $\ker(\Delta_1^*)$ has $\Bbbk$-dimension 1 so the image must have $\Bbbk$-dimension $6 - 1 = 5$.

For any morphism $g \in \mathrm{Hom}_{A^e}(A \otimes_{\Bbbk Q_0} \Bbbk\Gamma_1 \otimes_{\Bbbk Q_0} A, A)$, notice that all $\alpha \in \Gamma_1$ yields $g(1 \otimes \alpha \otimes 1) = k\alpha$ for $k \in \Bbbk$, by A^e-linearity. It follows that

$$\begin{aligned}\Delta_2^*(g)(1 \otimes ab \otimes 1) &= g\Delta_2(1 \otimes ab \otimes 1)\\ &= ag(1 \otimes b \otimes 1) + g(1 \otimes a \otimes 1)b\\ &= k_1ab + k_2ab = 0 + 0,\end{aligned}$$

for some k_1 and $k_2 \in \Bbbk$. The same holds for either of the two other paths in Γ_2, and thereby linearity ensures it holds for any element in $A \otimes_{\Bbbk Q_0} \Bbbk\Gamma_2 \otimes_{\Bbbk Q_0} A$. Hence, $\Delta_2^* = 0$, i.e. $\ker(\Delta_2^*) = \Bbbk^6$, so $HH^1(A) \cong \Bbbk^6/\Bbbk^5 \cong \Bbbk$.

On the other hand, it can be easily shown by counterexample that Δ_3^* and Δ_4^* are both non-zero. By $\Bbbk$-linearity, we can conclude $\mathrm{im}(\Delta_3^*) \cong \mathrm{im}(\Delta_4^*) \cong \Bbbk$, meaning Δ_3^* is injective and $\ker(\Delta_4^*)$ must be isomorphic to $\Bbbk$. Thus, $HH^n(A) = 0$ for $n > 1$ and the only non-zero Hochschild cohomology groups are

$$HH^0(A) \cong \Bbbk \quad \text{and} \quad HH^1(A) \cong \Bbbk.$$

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